Jia-Wei Liu

Curriculum Vitae

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Education

Sept. 2018 – July 2020 **Tsinghua University**, *Master of Science*, Department of Engineering Mechanics, Institute of Dynamics and Control, Advisor: Prof. Ge-Xue Ren, Overall GPA: 3.70/4.0.

Sept. 2014 – July 2018 **Northwestern Polytechnical University**, *Bachelor of Engineering*, Honors College. Overall GPA: 3.80/4.0, 91.84/100, Rank: 2/145, Major GPA: 3.88/4.0, 93.96/100.

Standardized Test

May 2019 TOEFL iBT: R28/L30/S21/W27, Total: 106.

Jan. 2019 GRE General: Verbal: 158(80%), Quantitative: 169(96%), AW: 4.0(59%), Total: 327+4.0.

Publications

Aug. 2019 **Jia-Wei Liu**, Jia-Peng Liu, Xuan-Bo Shu, Aki Mikkola, Ge-Xue Ren, "An efficient multibody dynamic model of three-dimensional meshing contacts in helical gear-shaft system and its solution", Mechanism and Machine Theory (SCI source, Q1), Accepted.

Feb. 2019 **Jia-Wei Liu**, Jia-Peng Liu, Jia-Lun Yao, Ge-Xue Ren, "Dynamic characteristics of helical gear based on Arbitrary Lagrangian Eulerian formulation", in: ECCOMAS 2019 Multibody Dynamic Conference, Duisburg, Germany, 2019, Accepted.

Research Interests

Dynamics and control, Robotics, Computational mechanics, Mechanical system.

Research Experience

Oct. 2018 – Present **Model order reduction method for simulation of planetary gear set based on Arbitrary Lagrangian Eulerian (ALE) formulation**, Research leader, Tsinghua University.

Proposing a novel ALE planetary gear set to reduce the degrees of freedom (DOFs) of the system from 10^6 to $10^3 \sim 10^4$. By splitting the planet gear into two ALE gears, each ALE planet gear only contacts with the ring or sun gear. We have verified the accuracy and efficiency of this method and the paper on this work is under preparation.

Feb. 2019 – Aug. 2019 **An efficient multibody dynamic modeling method of three-dimensional (3D) contacts in helical gear-shaft system**, Research leader, Tsinghua University.

Proposed the ALE helical gear, where low-frequency approximation (LFA) technique and ALE formulation were adopted to reduce the system DOFs from 10^6 to $10^3 \sim 10^4$. Then, we simplified the inertial terms in dynamic equations and adopted a high-efficiency contact algorithm based on hierarchical bounding box tree (OBB-tree). Finally, 3D Timoshenko beam elements were adopted to discretize the shaft. The proposed method was implemented using C++ and one order of magnitude acceleration of simulation time was achieved with high accuracy compared with ABAQUS.

June 2018 – Feb. 2019 **Development of evaluation method for sliding wear by multibody dynamics**, Project leader, Tsinghua University, Partner: Mitsubishi Heavy Industries, Japan.

Established a general low-frequency approximated model using C++, which was successfully applied in the variable-geometry turbocharger (VGT) in this project to improve the computational efficiency by five times. Adopted the hierarchical OBB-tree method for rapid contact detection in VGT.

Jan. 2018 – Oct. 2018 **Research on dynamic characteristics of helical gears and bevel gears based on ALE formulation**, Research leader, Tsinghua University.
Established the FE models of four pairs of helical gears with different helix angles and two pairs of bevel gears using Solidworks and Hypermesh. Developed the corresponding ALE gears for higher efficiency. Analyzed the dynamic transmission characteristics with correlation to three dimensional contacts and helix angles, for which an algorithm to compute contact lines was proposed.

Work Experience

Jan. 2018 – Present **Research Assistant**, Multibody Dynamics Laboratory, Tsinghua University.
Contributed to the research of multibody dynamics of gear system based on the ALE formulation, including the helical gear pair, bevel gear pair, helical gear-shaft system and planetary gear set.

Sept. 2018 – Jan. 2019 **Teaching Assistant**, Theoretical Mechanics, Tsinghua University.
Provided 2 lectures on laboratory sessions for introductions of the theoretical mechanics. Offered homework correction twice a week and presented 5 Q&A sessions. Invigilated and graded on 3 exams.

Volunteers & Activities

Sept. 2018 – Jan. 2019 **Principle Member**, League Graduate, Tsinghua University.
Assisted in the design of a memorial video and collaborated on encouraging students participation.

Sept. 2018 – Jan. 2019 **Principle Member**, Graduate Student Union, Tsinghua University.
Served as volunteer in the opening ceremony. Assisted in organizing academic lectures.

Apr. 2016 – June 2016 **Team Leader**, Project Management, Northwestern Polytechnical University.
Worked with teachers from Netherlands and led a team to organize the route plannings in Africa.

Sept. 2015 – Sept. 2016 **Principle Member**, Enactus Practice Center, Northwestern Polytechnical University.
Collected and analyzed data for the water-shortage region and collaborated on inventing our first water filter equipment, which adopted soil infiltration and was about 5 times cheaper than the market.

Sept. 2014 – July 2018 **Class Leader**, Honors College, Northwestern Polytechnical University.
Offered supports for students' study and organized dozens of Q&A sessions and study lectures.

Honors & Awards

Jan. 2018 Scholarship of Innovation and Entrepreneurship in Honors College.
June 2017 Second Prize of C++ Programming in Northwestern Polytechnical University.
Mar. 2017 Third Prize of National English Contest for College Student.
2016 – 2017 Scholarship of First Prize in Honors College.
2016 – 2017 Scholarship of Outstanding Student in Northwestern Polytechnical University.
May 2016 Second Prize of Mathematical Modeling in Northwestern Polytechnical University.
2015 – 2016 Scholarship of First Prize in Honors College.
2015 – 2016 Scholarship of Outstanding Student in Northwestern Polytechnical University.
May 2015 Second Prize of Mathematical Modeling in Northwestern Polytechnical University.
2014 – 2015 Scholarship of First Prize in Honors College.
2014 – 2015 Scholarship of Outstanding Student in Northwestern Polytechnical University.
2014 – 2015 National Scholarship (Awarded to top 1% students in Honors College).

Skills & Abilities

Proficient in C/C++ Programming, Matlab, ABAQUS, Hypermesh, Solidworks, Adobe Illustrator, Latex(Lyx), MS Powerpoint, MS Word, MS Excel.